**Title: Large Language Models Acting as Multidisciplinary Tumor Boards in Treatment Allocation for Hepatocellular Carcinoma**

**Table of Contents**


1. **Supplementary Tables S1 and S2**
2. **LLM Prompt**
3. **Standardized patient template**

2. **Supplementary Table S1. Post-hoc pairwise exact McNemar tests, by recommendation position**

| Pairwise comparison | n (A yes, B no) | n (A no, B yes) | p (exact McNemar) | p (Holm-adjusted) |
|---|---|---|---|---|
| **1st recommendation (Cochran's Q, p = 0.0007)** | | | | |
| ChatGPT 5 vs Gemini Flash | 28 | 10 | 0.005 | **0.031** |
| ChatGPT 5 vs Tumor Board ChatGPT | 10 | 7 | 0.629 | 1.000 |
| ChatGPT 5 vs Gemini 2.5 Pro | 5 | 4 | 1.000 | 1.000 |
| Gemini Flash vs Tumor Board ChatGPT | 8 | 23 | 0.011 | **0.043** |
| Gemini Flash vs Gemini 2.5 Pro | 10 | 27 | 0.008 | **0.038** |
| Tumor Board ChatGPT vs Gemini 2.5 Pro | 8 | 10 | 0.815 | 1.000 |
| **2nd recommendation (Cochran's Q, p < 0.0001)** | | | | |
| ChatGPT 5 vs Gemini Flash | 19 | 3 | <0.001 | **0.004** |
| ChatGPT 5 vs Tumor Board ChatGPT | 6 | 6 | 1.000 | 1.000 |
| ChatGPT 5 vs Gemini 2.5 Pro | 4 | 6 | 0.754 | 1.000 |
| Gemini Flash vs Tumor Board ChatGPT | 5 | 21 | 0.002 | **0.010** |
| Gemini Flash vs Gemini 2.5 Pro | 2 | 20 | <0.001 | **<0.001** |
| Tumor Board ChatGPT vs Gemini 2.5 Pro | 4 | 6 | 0.754 | 1.000 |
| **3rd recommendation (Cochran's Q, p = 0.048)** | | | | |
| ChatGPT 5 vs Gemini Flash | 14 | 5 | 0.064 | 0.381 |
| ChatGPT 5 vs Tumor Board ChatGPT | 4 | 4 | 1.000 | 1.000 |
| ChatGPT 5 vs Gemini 2.5 Pro | 6 | 3 | 0.508 | 1.000 |

| Pairwise comparison | n (A yes, B no) | n (A no, B yes) | p (exact McNemar) | p (Holm-adjusted) |
|---|---|---|---|---|
| Gemini Flash vs Tumor Board ChatGPT | 5 | 14 | 0.064 | 0.381 |
| Gemini Flash vs Gemini 2.5 Pro | 7 | 13 | 0.263 | 1.000 |
| Tumor Board ChatGPT vs Gemini 2.5 Pro | 5 | 2 | 0.453 | 1.000 |

n( A yes, B no) and n (A no, B yes) are the discordant pair counts, indicate the direction of disagreement between the two models.

**Supplementary Table S2**. Sensitivity and precision reported per treatment categories (with a sufficient number of cases > 9 true cases)

| Category (small n) | n | ChatGPT Sens/Prec | TB ChatGPT Sens/Prec | Gemini Flash Sens/Prec | Gemini Pro Sens/Prec |
|---|---|---|---|---|---|
| **BSC** | 7 | 0.714 / 1.000 | 0.714 / 0.833 | 0.429 / 0.750 | 0.714 / 1.000 |
| **LT, bridge with TACE** | 7 | 1.000 / 1.000 | 0.714 / 0.833 | 0.714 / 1.000 | 0.857 / 1.000 |
| **Liver transplantation** | 4 | 0.750 / 0.300 | 1.000 / 0.444 | 0.500 / 0.286 | 0.750 / 0.429 |
| **TARE** | 2 | 0.000 / 0.000 | 0.000 / 0.000 | 0.000 / NaN | 0.000 / 0.000 |
| **LT, bridge with ablation** | 1 | 0.000 / NaN | 0.000 / NaN | 0.000 / NaN | 0.000 / NaN |
| **Systemic therapy until LT** | 1 | 1.000 / 1.000 | 0.000 / NaN | 0.000 / NaN | 1.000 / 1.000 |

## 1. LLM prompt

Act as a multidisciplinary Tumor Board within an Institute of Gastroenterology, in a tertiary hospital setting. The Tumor Board consists of: hepato-biliary and pancreatic surgeon, radiologist, interventional radiologist, radiotherapist, gastroenterologist, hepatologist, and oncologist.

You will be provided with a patient case. Follow this process:

1. Restate the case concisely.
2. Present each specialist's reasoning separately.
3. Apply the relevant guidelines on HCC treatment: EASL 2018 and AASLD 2018. For scenarios involving tumor ablation or chemoembolization, also apply CIRSE guidelines (2019 and 2021).
4. When guidelines differ, summarize both views and indicate the choice aligned with our Institute's policy.
5. Apply these institutional protocols:
   - Liver transplantation unavailable before September 2024, available afterwards.
   - TACE used with curative intent for tumors <3 cm in patients with contraindications for surgery and ablation.
6. Provide the consensus treatment recommendation, explicitly stating rationale.
7. Suggest alternative options.

## 2. Patient template

**FORM FOR ALLOCATION TO SPECIFIC TREATMENT**

**FOR PATIENTS WITH HEPATOCELLULAR CARCINOMA No. .....**

aged

signed the informed consent form for treatment NO YES, on __/__/____

**Current clinical and laboratory data**

| | | | | | | |
|---|---|---|---|---|---|---|
| Height | Weight | ascites | encephalopathy | | | **ECOG** |
| LSM | HVPG | Esophageal varices | **Child-Pugh** | | **MELD** | **AFP** |
| Hb | WBC | PLT | INR | Albumin | Total bilirubin | Direct bilirubin |
| ALP | GGT | AST | ALT | Creatinine | | *NYHA class* |
| HBsAg | antiHCV | antiHDV | | | | *LVEF* |

**Medical history**

| | | |
|---|---|---|
| ***liver disease*** | date of cirrhosis / nodule(s) diagnosis | __/__/____ // __/__/____ |
| | previous antiviral treatment | NO YES drug _____ period _____ |
| | previous treatment for liver nodule(s) | NO YES modality _____ date _____ |
| | hepatic decompensation | NO YES type _____ date _____ |
| ***associated conditions*** | | |
| ***current treatments*** | | |

**Imaging**

| | **abdominal US (including CEUS)** | **contrast-enhanced CT** | **MRI** |
|---|---|---|---|
| ***date*** | | | |
| ***no. of nodules*** | | | |
| ***diameter (largest)*** | | | |
| ***segment*** | | | |
| ***LIRADS*** | | | |
| ***tumor volume / remaining liver volume*** | | | |
| ***PV / HV thrombosis*** | NO YES Ben Mal | NO YES Ben Mal | NO YES Ben Mal |
| ***lymph nodes*** | NO YES group_____ | NO YES group_____ | NO YES group_____ |
| ***extrahepatic disease*** | NO YES site _____ | NO YES site _____ | NO YES site _____ |

**Final diagnosis**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| **BCLC stage** | | ***CLIP score*** | | ***HAP score*** | | ***ART score (if applicable)*** | |

I, the attending physician _____, certify the accuracy and completeness of the information provided above.

(signature and stamp)

After reviewing the above data, the multidisciplinary committee of IRGH, convened on __/__/____, proposes the following therapeutic approach

---

- first-line treatment

| | | | | | |
|---|---|---|---|---|---|
| **LIVER TRANSPLANTATION** | ❑ | | **CHEMOEMBOLIZATION (TACE)** | ❑ | **CLINICAL TRIALS** |
| **SURGICAL RESECTION** | ❑ D | ❑ L | **COMBINED THERAPY (TACE+RFA/MWA)** | ❑ | |
| | | | **SYSTEMIC THERAPY** | ❑ | |
| **ABLATION (RFA/MWA)** | ❑ pc | ❑io ❑ L | **SYSTEMIC THERAPY** | ❑ | **SYMPTOMATIC TREATMENT (BSC)** |

- second-line treatment

| | | | | | |
|---|---|---|---|---|---|
| **LIVER TRANSPLANTATION** | ❑ | | **CHEMOEMBOLIZATION (TACE)** | ❑ | **CLINICAL TRIALS** |
| **SURGICAL RESECTION** | ❑ D | ❑ L | **COMBINED THERAPY (TACE+RFA/MWA)** | ❑ | |
| | | | **SYSTEMIC THERAPY** | ❑ | |
| **ABLATION (RFA/MWA)** | ❑ pc | ❑io ❑ L | **SYSTEMIC THERAPY** | ❑ | **SYMPTOMATIC TREATMENT (BSC)** |